

Computing limits of sequences



Outline

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Introduction: Computing limits of sequences

- ▶ Verifying a limit with the definition is laborious: we need rules to **compute** limits
- ▶ In this section we see how the limit behaves with respect to the **algebraic operations** (algebra of limits) and with respect to the **order** (sign permanence, comparison)
- ▶ We then extend the operations to the symbols $\pm\infty$ (**partial arithmetization**) and identify the **indeterminate forms**
- ▶ Finally we learn to handle limits of powers $a_n^{b_n}$ by taking the **logarithm**

- ▶ The proofs of the **basic theorems** on computing limits are based on the **definition of limit**, on the **use of inequalities**, and on the use of **properties that are eventually true**.
- ▶ In particular, these theorems illustrate the relation between the limit operation and the algebraic structures¹ and the order structures² present in $\mathbb{R}$.

¹A set on which an operation is defined is called an algebraic structure

²A set is endowed with an order structure if an order relation is defined on it

Computing limits of sequences

We will simply write $a_n \rightarrow \ell$ to mean $a_n \rightarrow \ell$ for $n \rightarrow +\infty$

Properties of the limit operation with respect to algebraic operations.

Theorem 1: Algebra of limits, case of finite limits

Hypotheses:

$$1. \ a_n \rightarrow \ell_a \in \mathbb{R} \qquad 2. \ b_n \rightarrow \ell_b \in \mathbb{R}.$$

Claim:

$$\begin{array}{ll} 1. \ a_n \pm b_n \rightarrow \ell_a \pm \ell_b & 2. \ \frac{a_n}{b_n} \rightarrow \frac{\ell_a}{\ell_b} \quad (b_n, \ell_b \neq 0, \text{ eventually}) \\ 3. \ a_n b_n \rightarrow \ell_a \ell_b & 4. \ a_n^{b_n} \rightarrow \ell_a^{\ell_b} \quad (a_n, \ell_a > 0, \text{ eventually}). \end{array}$$

Proof (1/2)

We prove that:

$$a_n \rightarrow \ell_a \in \mathbb{R}, \quad b_n \rightarrow \ell_b \in \mathbb{R} \Rightarrow a_n + b_n \rightarrow \ell_a + \ell_b$$

We consider

$$|(a_n + b_n) - (\ell_a + \ell_b)| = |(a_n - \ell_a) + (b_n - \ell_b)| \leq |a_n - \ell_a| + |b_n - \ell_b| \quad (1)$$

by the triangle inequality. Since by hypothesis $a_n \rightarrow \ell_a$ and $b_n \rightarrow \ell_b$, we have that

$$|a_n - \ell_a| < \varepsilon_a \quad \text{and} \quad |b_n - \ell_b| < \varepsilon_b \quad (\text{eventually})$$

Proof (2/2)

for every $\varepsilon_a > 0$ and $\varepsilon_b > 0$. Bounding from above the terms on the right-hand side of (1) we conclude that

$$|(a_n + b_n) - (\ell_a + \ell_b)| < \underbrace{2 \max\{\varepsilon_a, \varepsilon_b\}}_{=\tilde{\varepsilon} \text{ and } >0}$$

Since $\tilde{\varepsilon}$ is arbitrary, the claim follows. □

Proof (1/2)

We prove that:

$$a_n \rightarrow \ell_a \in \mathbb{R}, b_n \rightarrow \ell_b \in \mathbb{R} \Rightarrow a_n - b_n \rightarrow \ell_a - \ell_b$$

For every $\varepsilon_a > 0$ and $\varepsilon_b > 0$, we have:

$$|(a_n - b_n) - (\ell_a - \ell_b)| = |(a_n - \ell_a) + (\ell_b - b_n)| \leq \underbrace{|a_n - \ell_a|}_{< \varepsilon_a} + \underbrace{|\ell_b - b_n|}_{= |b_n - \ell_b| < \varepsilon_b}$$

Hence

$$|(a_n - b_n) - (\ell_a - \ell_b)| < \underbrace{2 \max\{\varepsilon_a, \varepsilon_b\}}_{=\tilde{\varepsilon} \text{ and } >0}$$

Proof (2/2)

Since $\tilde{\varepsilon}$ is arbitrary, the claim follows.



Proof (1/3)

We prove that:

$$a_n \rightarrow \ell_a \in \mathbb{R}, \quad b_n \rightarrow \ell_b \in \mathbb{R} \Rightarrow a_n b_n \rightarrow \ell_a \ell_b$$

We consider

$$\begin{aligned} |(a_n b_n) - (\ell_a \ell_b)| &= |a_n b_n - a_n \ell_b + a_n \ell_b - \ell_a \ell_b| \\ &= |a_n (b_n - \ell_b) + \ell_b (a_n - \ell_a)| \leq |a_n (b_n - \ell_b)| + |\ell_b (a_n - \ell_a)| \\ &= |a_n| |b_n - \ell_b| + |\ell_b| |a_n - \ell_a| \end{aligned}$$

Proof (2/3)

by the triangle inequality and the properties of the absolute value.
Hence

$$|(a_n b_n) - (\ell_a \ell_b)| \leq |a_n| |b_n - \ell_b| + |\ell_b| |a_n - \ell_a| \quad (2)$$

Since by hypothesis $a_n \rightarrow \ell_a$ and $b_n \rightarrow \ell_b$, for every $\varepsilon_a > 0$ and $\varepsilon_b > 0$ we have

$$|a_n - \ell_a| < \varepsilon_a \quad \text{and} \quad |b_n - \ell_b| < \varepsilon_b \quad (\text{eventually}).$$

Proof (3/3)

Moreover, since

$$|a_n - \ell_a| \geq |a_n| - |\ell_a| \quad \text{we have} \quad |a_n| < |\ell_a| + \varepsilon_a, \quad \text{eventually.}$$

Therefore, bounding from above the terms on the right-hand side of (2), we conclude that

$$|(a_n b_n) - (\ell_a \ell_b)| < (|\ell_a| + \varepsilon_a) \varepsilon_b + |\ell_b| \varepsilon_a = \underbrace{|\ell_a| \varepsilon_b + |\ell_b| \varepsilon_a + \varepsilon_a \cdot \varepsilon_b}_{=\tilde{\varepsilon} \text{ and } >0}$$

Since $\tilde{\varepsilon}$ is arbitrary, the claim follows. □

- Moreover, the **limit operation preserves the ordering**

Theorem 2: Sign-preservation, first form

Hypotheses:

$$1. \ a_n \rightarrow \ell_a \qquad 2. \ \ell_a \leq 0.$$

Claim:

$$a_n \leq 0, \text{ eventually.}$$

Proof (1/2)

We consider the case $\ell_a > 0$. By definition of limit we have that

$$|a_n - \ell_a| < \varepsilon, \quad \text{eventually,}$$

for every $\varepsilon > 0$, which we rewrite in the form:

$$\ell_a - \varepsilon < a_n < \ell_a + \varepsilon, \quad \text{eventually.}$$

Proof (2/2)

Since $\ell_a > 0$, we can choose $\varepsilon > 0$ such that $\ell_a - \varepsilon > 0$; then the inequality

$$0 < \ell_a - \varepsilon < a_n$$

shows that $a_n > 0$, eventually.

The case $\ell_a < 0$ is proved analogously.



Theorem 3: Permanence of sign (generalized form)

Hypotheses:

$$1. a_n \rightarrow \ell_a \in \mathbb{R} \quad 2. \lambda \in \mathbb{R} \quad 3. \ell_a > \lambda.$$

Claim:

$$a_n > \lambda, \text{ eventually.}$$

Proof

Since $\ell_a > \lambda$, the number $\varepsilon = \ell_a - \lambda$ is strictly positive. Applying the definition of limit with precisely this value of ε , we have, eventually:

$$\underbrace{\ell_a - (\ell_a - \lambda)}_{=\lambda} < a_n < \ell_a + (\ell_a - \lambda)$$

and therefore $a_n > \lambda$, eventually. □

Computing limits of sequences

- ▶ Analogously: if $a_n \rightarrow \ell_a \in \mathbb{R}$ and $\ell_a < \lambda$, then $a_n < \lambda$ eventually (choose $\varepsilon = \lambda - \ell_a > 0$)
- ▶ With $\lambda = 0$ we recover the permanence of sign theorem, 1st form
- ▶ We can now prove the case of the **quotient** in the theorem on the algebra of limits

Proof (1/3)

We first prove that:

$$b_n \rightarrow \ell_b \in \mathbb{R}, \ell_b \neq 0 \Rightarrow \frac{1}{b_n} \rightarrow \frac{1}{\ell_b}$$

The case of the quotient then follows from the case of the product, already proved, applied to the sequences $\{a_n\}$ and $\left\{\frac{1}{b_n}\right\}$:

$$\frac{a_n}{b_n} = a_n \frac{1}{b_n} \rightarrow \ell_a \frac{1}{\ell_b} = \frac{\ell_a}{\ell_b}$$

Proof (2/3)

Suppose $\ell_b > 0$ (if $\ell_b < 0$ one argues in the same way on the sequence $\{-b_n\}$). By the permanence of sign theorem (generalized form), applied with $\lambda = \frac{\ell_b}{2} < \ell_b$, we have

$$b_n > \frac{\ell_b}{2} > 0, \quad \text{eventually}$$

in particular $b_n \neq 0$ eventually and the quotient $\frac{1}{b_n}$ is well defined.

Proof (3/3)

For every $\varepsilon > 0$ we have $|b_n - \ell_b| < \varepsilon$ eventually, hence, eventually:

$$\begin{aligned} \left| \frac{1}{b_n} - \frac{1}{\ell_b} \right| &= \left| \frac{\ell_b - b_n}{b_n \ell_b} \right| = \frac{|b_n - \ell_b|}{|b_n| |\ell_b|} \\ &< \frac{2}{\ell_b |\ell_b|} |b_n - \ell_b| < \underbrace{\frac{2}{\ell_b^2} \varepsilon}_{=\tilde{\varepsilon} \text{ and } >0} \end{aligned}$$

where we used $|b_n| > \frac{\ell_b}{2}$. Since $\tilde{\varepsilon}$ is arbitrary, the claim follows.



Theorem 4: Sign-preservation, second form (part I)

Hypotheses:

1. $a_n \rightarrow \ell_a \in \mathbb{R}$
2. $a_n \geq 0$, eventually.

Claim:

$$\ell_a \geq 0.$$

Proof

It follows from the previous theorem. Indeed, if by contradiction we had $\ell_a < 0$, from the previous theorem we would have $a_n < 0$ eventually, which is incompatible with the hypothesis that $a_n \geq 0$ eventually.

This case cannot occur, i.e., the opposite holds, which is the claim of the theorem. □

Theorem 5: Sign-preservation, second form (part II)

Hypotheses:

1. $a_n \rightarrow \ell_a \in \mathbb{R}$ 2. $b_n \rightarrow \ell_b \in \mathbb{R}$ 3. $a_n \geq b_n$, eventually.

Claim:

$$\ell_a \geq \ell_b.$$

Proof

We consider the sequence $a_n - b_n$; by the theorem on the algebra of limits we have

$$a_n - b_n \rightarrow \ell_a - \ell_b.$$

Since by hypothesis

$$a_n \geq b_n \quad \text{and hence} \quad a_n - b_n \geq 0,$$

by the sign-preservation theorem, second form (part I), applied to the sequence $a_n - b_n$ we have

$$\ell_a - \ell_b \geq 0 \quad \text{and hence} \quad \ell_a \geq \ell_b.$$



- ▶ This theorem tells us that in an inequality between two sequences we can pass to the limit on both sides, keeping the “ $\leq$ ” or the “ $\geq$ ”.
- ▶ Note, instead, that in general strict inequalities “ $<$ ” and “ $>$ ” are not preserved when passing to the limit.

Example: Passing to the limit with strict inequalities

For example, even if the a_n are strictly positive, their limit ℓ_a is positive or zero, as shown by the simple example $\frac{1}{n} \rightarrow 0$.

Theorem 6: Comparison (squeeze) theorem

Hypotheses:

1. $a_n \rightarrow \ell, c_n \rightarrow \ell$ and $\ell \in \mathbb{R}$, 2. $a_n \leq b_n \leq c_n$, eventually.

Claim:

$$b_n \rightarrow \ell.$$

Proof (1/2)

By definition of limit we have, eventually, that

$$\ell - \varepsilon_a < a_n < \ell + \varepsilon_a \quad \text{and} \quad \ell - \varepsilon_c < c_n < \ell + \varepsilon_c$$

for every $\varepsilon_a > 0$ and $\varepsilon_c > 0$. Hence:

$$\ell - \max\{\varepsilon_a, \varepsilon_c\} < a_n < \ell + \max\{\varepsilon_a, \varepsilon_c\}$$

$$\text{and} \quad \ell - \max\{\varepsilon_a, \varepsilon_c\} < c_n < \ell + \max\{\varepsilon_a, \varepsilon_c\}$$

Proof (2/2)

From the hypotheses of the theorem we therefore have, eventually, that

$$\ell - \max\{\varepsilon_a, \varepsilon_c\} < a_n \leq b_n \leq c_n < \ell + \max\{\varepsilon_a, \varepsilon_c\}$$

But then, eventually, we have

$$\underbrace{\ell - \max\{\varepsilon_a, \varepsilon_c\}}_{=\tilde{\varepsilon} \text{ and } >0} < b_n < \ell + \underbrace{\max\{\varepsilon_a, \varepsilon_c\}}_{=\tilde{\varepsilon} \text{ and } >0}.$$

Since $\tilde{\varepsilon}$ is arbitrary, the claim follows. □

Theorem 7: Comparison theorem for divergent sequences

Hypotheses:

$$1. \ a_n \rightarrow +\infty, \quad 2. \ a_n \leq b_n, \text{ eventually.}$$

Claim:

$$b_n \rightarrow +\infty.$$

Proof

Since $a_n \rightarrow +\infty$, for every $M > 0$ we have, eventually,

$$a_n > M$$

By hypothesis we also have $a_n \leq b_n$, eventually. Both properties hold eventually, hence, eventually, we have

$$b_n \geq a_n > M$$

Since $M > 0$ is arbitrary, we conclude that $b_n \rightarrow +\infty$. □

- ▶ Analogously one proves that, if $a_n \rightarrow -\infty$ and $a_n \geq b_n$ eventually, then $b_n \rightarrow -\infty$
- ▶ Frequently used special cases of the comparison theorem are expressed by the following corollaries, which are very useful when studying the product of an oscillating (but bounded) sequence and one that tends to zero

Corollary 1: Of the comparison theorem (part I)

Hypotheses:

1. $c_n \rightarrow 0$
2. $|b_n| \leq c_n$, eventually.

Claim:

$$b_n \rightarrow 0.$$

Proof

We know that eventually we have $-c_n \leq b_n \leq c_n$. Obviously

if $c_n \rightarrow 0$ then $-c_n \rightarrow 0$.

Hence by the comparison theorem (with $a_n = -c_n$ and $\ell = 0$) we have that $b_n \rightarrow 0$. □

Corollary 2: Of the comparison theorem (part II)

Hypotheses:

1. $c_n \rightarrow 0$ 2. $\{b_n\}$ is bounded (but not necessarily convergent).

Claim:

$$c_n b_n \rightarrow 0.$$

Proof

If $\{b_n\}$ is bounded, then $|b_n| \leq M$ for some $M > 0$ and for every $n \in \mathbb{N}$. We can therefore write

$$|b_n c_n| \leq M |c_n|.$$

Since

$$c_n \rightarrow 0 \quad \text{also} \quad M |c_n| \rightarrow 0,$$

by Corollary we conclude that $b_n c_n \rightarrow 0$.



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The product of an infinitesimal sequence and a bounded one is infinitesimal.

Example: Application of the corollary (1/2)

Consider the sequence given by a ratio of two expressions, each consisting of a sum of powers of n , such as:

$$n \mapsto \frac{n^{5/2} - 3n + 7}{n^3 + \sqrt{n} - 3n^2}$$

Factoring out the highest power both in the numerator and in the denominator we obtain:

$$\frac{n^{5/2} \left(1 - \frac{3}{n^{3/2}} + \frac{7}{n^{5/2}}\right)}{n^3 \left(1 + \frac{1}{n^{5/2}} - \frac{3}{n}\right)} = \frac{1}{\sqrt{n}} \frac{1 - \frac{3}{n^{3/2}} + \frac{7}{n^{5/2}}}{1 + \frac{1}{n^{5/2}} - \frac{3}{n}}$$

Example: Application of the corollary (2/2)

Now, by Theorem on the algebra of limits and knowing that negative powers of n tend to zero, we can state that:

$$1 - \underbrace{\frac{3}{n^{3/2}}}_{\rightarrow 0} + \underbrace{\frac{7}{n^{5/2}}}_{\rightarrow 0} \rightarrow 1, \quad 1 + \underbrace{\frac{1}{n^{5/2}}}_{\rightarrow 0} - \underbrace{\frac{3}{n}}_{\rightarrow 0} \rightarrow 1 \quad \text{and} \quad \left(\frac{1 - \frac{3}{n^{3/2}} + \frac{7}{n^{5/2}}}{1 + \frac{1}{n^{5/2}} - \frac{3}{n}} \right) \rightarrow 1$$

hence the last sequence is convergent and consequently bounded.
Now, by Corollary and since

$$\frac{1}{\sqrt{n}} \rightarrow 0 \quad \text{we have} \quad \frac{n^{5/2} - 3n + 7}{n^3 + \sqrt{n} - 3n^2} \rightarrow 0$$

Example: Application of the corollary (1/2)

The sequence

$$n \mapsto \frac{\sin n}{n}$$

is the product of two sequences

$$n \mapsto \frac{1}{n} \text{ (convergent, infinitesimal) and } n \mapsto \sin n \text{ (irregular),}$$

hence Theorem on the algebra of limits cannot be applied (the second limit does not exist).

Example: Application of the corollary (2/2)

- ▶ However, Corollary can be applied. The sequence $\{\frac{1}{n}\}$ is infinitesimal and, since $|\sin n| \leq 1$, the sequence $\{\sin n\}$ is bounded; therefore we have

$$\lim_{n \rightarrow +\infty} \frac{\sin n}{n} = 0$$

- So far we have seen theorems that work on pairs of sequences that are both convergent or at least bounded.

Sequences with limits $+\infty$ and $-\infty$

- Suppose, for example, that

$$a_n \rightarrow \ell_a \quad \text{and} \quad b_n \rightarrow +\infty$$

then it is easy (and intuitive) to see that

$$a_n + b_n \rightarrow +\infty$$

We will abbreviate this as follows:

$$\ell_a + \infty = +\infty$$

- Reasoning in a similar way we can summarize the rules for the limit of the sum (or difference) of two sequences, one or both of which are divergent.

Rules of partial arithmetization of the infinity symbol

Theorem 8: Partial arithmetization of the infinity symbol (addition)

Hypotheses:

$$1. a_n \rightarrow \ell_a \in \mathbb{R} \quad 2. b_n \rightarrow +\infty \quad 3. c_n \rightarrow +\infty.$$

Claim:

$$\begin{array}{ll} 1. a_n + b_n \rightarrow \ell_a + \infty = +\infty & 2. a_n - b_n \rightarrow \ell_a - \infty = -\infty \\ 3. b_n + c_n \rightarrow +\infty + \infty = +\infty & 4. -b_n - c_n \rightarrow -\infty - \infty = -\infty. \end{array}$$

Theorem 9: Partial arithmetization of the infinity symbol (product)

Hypotheses:

$$1. a_n \rightarrow \ell_a \in \mathbb{R} \quad 2. b_n \rightarrow 0^+ \text{ or } b_n \rightarrow 0^- \quad 3. c_n \rightarrow \infty.$$

Claim:

$$1. a_n c_n \rightarrow \ell_a \infty = \infty \quad (\ell_a \neq 0)$$

$$2. \frac{a_n}{b_n} \rightarrow \frac{\ell_a}{0} = \infty \quad (\ell_a \neq 0)$$

$$3. \frac{a_n}{c_n} \rightarrow \frac{\ell_a}{\infty} = 0.$$

- ▶ the **sign** of ∞ must be determined with the **usual rule of signs**
- ▶ in claim 2. it is understood, as usual, that $b_n \neq 0$ eventually, so that the quotient is well defined
- ▶ hypothesis 2., namely that $\{b_n\}$ tends to zero *from above* or *from below* (and hence that it eventually has constant sign), is **necessary**: with $a_n = 1$ and $b_n = \frac{(-1)^n}{n} \rightarrow 0$ we have

$$\frac{a_n}{b_n} = (-1)^n n$$

which is an irregular sequence (it tends neither to $+\infty$ nor to $-\infty$)

Example: Rule of signs

We have:

- ▶ if $a_n \rightarrow \ell_a \in \mathbb{R}, \ell_a > 0$ and $b_n \rightarrow 0^+$ then $\frac{a_n}{b_n} \rightarrow +\infty$
- ▶ if $a_n \rightarrow \ell_a \in \mathbb{R}, \ell_a < 0$ and $b_n \rightarrow 0^-$ then $\frac{a_n}{b_n} \rightarrow +\infty$
- ▶ if $a_n \rightarrow \ell_a \in \mathbb{R}, \ell_a > 0$ and $b_n \rightarrow 0^-$ then $\frac{a_n}{b_n} \rightarrow -\infty$
- ▶ if $a_n \rightarrow \ell_a \in \mathbb{R}, \ell_a < 0$ and $b_n \rightarrow 0^+$ then $\frac{a_n}{b_n} \rightarrow -\infty$

Therefore, in order to apply the rules of partial arithmetization of the infinity symbol, it is necessary to determine whether b_n tends to zero from above or from below.

Proof (1/2)

We prove that:

$$a_n \rightarrow \ell_a \in \mathbb{R}, \quad c_n \rightarrow +\infty \quad \Rightarrow \quad \frac{a_n}{c_n} \rightarrow 0.$$

For every $\varepsilon > 0$, since $a_n \rightarrow \ell_a$, eventually we have

$$\underbrace{|a_n - \ell_a|}_{\geq |a_n| - |\ell_a|} < \varepsilon \quad \text{hence} \quad |a_n| < |\ell_a| + \varepsilon.$$

Proof (2/2)

Moreover, since $c_n \rightarrow +\infty$, eventually we have

$$c_n > \frac{1}{\varepsilon}$$

It follows that, eventually, we have

$$\frac{|a_n|}{|c_n|} < \frac{|\ell_a| + \varepsilon}{\frac{1}{\varepsilon}} \quad \text{and hence} \quad \left| \frac{a_n}{c_n} \right| < \varepsilon (|\ell_a| + \varepsilon) = \underbrace{|\ell_a| \varepsilon + \varepsilon^2}_{=\tilde{\varepsilon} \text{ and } >0}$$

Since $\tilde{\varepsilon}$ is arbitrary, the claim follows. □

Computing limits of sequences

The four missing operations:

$$+\infty - \infty, \quad 0 \cdot \infty, \quad \frac{0}{0} \quad \text{and} \quad \frac{\infty}{\infty}$$

are called **indeterminate forms**, since no rule can be established a priori to determine their result.

Example: Resolving indeterminate forms $+\infty - \infty$ (1/3)

Consider the sequence

$$n \mapsto \sqrt{n+1} - \sqrt{n-1}$$

We have the difference of two sequences:

$$n \mapsto \sqrt{n+1} \quad \text{with} \quad \lim_{n \rightarrow +\infty} \sqrt{n+1} = +\infty$$

$$n \mapsto \sqrt{n-1} \quad \text{with} \quad \lim_{n \rightarrow +\infty} \sqrt{n-1} = +\infty$$

Example: Resolving indeterminate forms $+\infty - \infty$ (2/3)

Hence we fall into the indeterminate form $+\infty - \infty$.

Multiplying and dividing by $\sqrt{n+1} + \sqrt{n-1}$ we obtain

$$\frac{(\sqrt{n+1} - \sqrt{n-1})(\sqrt{n+1} + \sqrt{n-1})}{\sqrt{n+1} + \sqrt{n-1}} = \frac{(\sqrt{n+1})^2 - (\sqrt{n-1})^2}{\sqrt{n+1} + \sqrt{n-1}} = \frac{2}{\sqrt{n+1} + \sqrt{n-1}}$$

recalling that $a^2 - b^2 = (a - b)(a + b)$. Now, considering the sequence in the denominator, we have

$$\lim_{n \rightarrow +\infty} \sqrt{n+1} + \sqrt{n-1} = +\infty + \infty = +\infty$$

Example: Resolving indeterminate forms $+\infty - \infty$ (3/3)

Using the rule

$$a_n \rightarrow \ell_a \in \mathbb{R}, \quad c_n \rightarrow \infty \quad \Rightarrow \quad \frac{a_n}{c_n} \rightarrow 0$$

of the theorem on the partial arithmetization of the infinity symbol (product), we have:

$$\lim_{n \rightarrow +\infty} \sqrt{n+1} - \sqrt{n-1} = \lim_{n \rightarrow +\infty} \frac{2}{\sqrt{n+1} + \sqrt{n-1}} = \frac{2}{+\infty} = 0$$

- Limits of sequences of the form:

$$\left\{ a_n^{b_n} \right\}$$

can be handled by considering the sequence of their logarithms, taking for simplicity the base e .

Computing limits of sequences

Given a sequence $\{a_n^{b_n}\}$, we have that:

$$\text{if } b_n \log a_n \rightarrow \ell \text{ then } a_n^{b_n} \rightarrow e^\ell$$

$$\text{if } b_n \log a_n \rightarrow +\infty \text{ then } a_n^{b_n} \rightarrow +\infty$$

$$\text{if } b_n \log a_n \rightarrow 0 \text{ then } a_n^{b_n} \rightarrow 1$$

$$\text{if } b_n \log a_n \rightarrow -\infty \text{ then } a_n^{b_n} \rightarrow 0$$

If the sequence $\{b_n \log a_n\}$ is indeterminate (has no limit), then $\{a_n^{b_n}\}$ is indeterminate as well.

Example: Computing limits by taking logarithms (1/2)

$$\lim_{n \rightarrow +\infty} (3n)^{(-3n^2+7)}$$

taking logarithms we have

$$\underbrace{(-3n^2+7)}_{b_n} \log \overbrace{3n}^{a_n}$$

since

$$\lim_{n \rightarrow +\infty} (-3n^2+7) \log(3n) = -\infty \cdot +\infty = -\infty$$

Example: Computing limits by taking logarithms (2/2)

then

$$\lim_{n \rightarrow +\infty} (3n)^{(-3n^2+7)} = 0$$

Example: Computing limits by taking logarithms (alternative method)

$$\lim_{n \rightarrow +\infty} (3n)^{(-3n^2+7)} = \lim_{n \rightarrow +\infty} e^{\log\left((3n)^{(-3n^2+7)}\right)} = \lim_{n \rightarrow +\infty} e^{(-3n^2+7) \log(3n)}$$

since

$$\lim_{n \rightarrow +\infty} (-3n^2 + 7) \log(3n) = -\infty \cdot +\infty = -\infty$$

then

$$\lim_{n \rightarrow +\infty} (3n)^{(-3n^2+7)} = e^{-\infty} = 0$$

Computing limits of sequences (1/2)

We also have the following **indeterminate forms**:

$$1^{\infty}, \quad 0^0, \quad \infty^0$$

Taking logarithms, they correspond to the indeterminate form

$$0 \cdot \infty$$

since:

$$\log(1^{\infty}) = \infty \log 1 = \infty \cdot 0$$

$$\log(0^0) = 0 \log 0 = 0 \cdot -\infty$$

Computing limits of sequences (2/2)

$$\log(+\infty^0) = 0 \quad \log(+\infty) = 0 \cdot +\infty$$

In the powers $\{a_n^{b_n}\}$ the base is always (eventually) positive, hence in the form ∞^0 the base is $+\infty$.

Computing limits of sequences

Limits of the form

$$0^{+\infty} \text{ and } 0^{-\infty}$$

are not indeterminate forms. We have:

$$0^{+\infty} = 0 \text{ and } 0^{-\infty} = +\infty.$$

since, taking logarithms, we have

$$\log(0^{+\infty}) = +\infty \cdot \log 0 = +\infty \cdot -\infty = -\infty \quad \text{and} \quad e^{-\infty} = 0$$

$$\log(0^{-\infty}) = -\infty \cdot \log 0 = -\infty \cdot -\infty = +\infty \quad \text{and} \quad e^{+\infty} = +\infty$$

Example: Limits of the forms $0^{+\infty}$ and $0^{-\infty}$ (1/3)

- ▶ Example of the case $0^{+\infty} = 0$:

$$\lim_{n \rightarrow +\infty} \left(\frac{1}{n}\right)^{\log n} = 0^{+\infty} = 0.$$

Indeed we have

$$\left(\frac{1}{n}\right)^{\log n} = e^{\log\left(\frac{1}{n}\right) \log n} = e^{\log n \log \frac{1}{n}}$$

hence

Example: Limits of the forms $0^{+\infty}$ and $0^{-\infty}$ (2/3)

$$\lim_{n \rightarrow +\infty} \left(\frac{1}{n}\right)^{\log n} = \lim_{n \rightarrow +\infty} e^{\overbrace{\log n}^{\rightarrow +\infty} \overbrace{\log \frac{1}{n}}^{\rightarrow -\infty}} = e^{-\infty} = 0.$$

► Example of the case $0^{-\infty} = +\infty$:

$$\lim_{n \rightarrow +\infty} \left(\frac{1}{n}\right)^{-\log n} = 0^{-\infty} = +\infty.$$

Indeed we have

$$\left(\frac{1}{n}\right)^{-\log n} = e^{\log\left(\frac{1}{n}\right)^{-\log n}} = e^{-\log n \log \frac{1}{n}}$$

Example: Limits of the forms $0^{+\infty}$ and $0^{-\infty}$ (3/3)

hence

$$\lim_{n \rightarrow +\infty} \left(\frac{1}{n}\right)^{-\log n} = \lim_{n \rightarrow +\infty} e^{\overbrace{-\log n}^{\rightarrow -\infty} \overbrace{\log \frac{1}{n}}^{\rightarrow -\infty}} = e^{+\infty} = +\infty.$$

Summary (1/2)

Algebra of limits (finite limits):

- ▶ If $a_n \rightarrow \ell_a$, $b_n \rightarrow \ell_b$: sums, differences, products, quotients ($\ell_b \neq 0$) and powers ($\ell_a > 0$) pass to the limit

Limits and order:

- ▶ **Sign permanence:** if $a_n \rightarrow \ell > 0$ then $a_n > 0$ eventually; if $a_n \geq 0$ eventually then $\ell \geq 0$
- ▶ **Comparison:** if $a_n \leq b_n \leq c_n$ and $a_n, c_n \rightarrow \ell$, then $b_n \rightarrow \ell$
- ▶ The product of an infinitesimal sequence and a bounded one is infinitesimal

Summary (2/2)

The infinity symbol:

- ▶ $\ell \pm \infty = \pm\infty$, $+\infty + \infty = +\infty$, $\ell \cdot \infty = \infty$ ($\ell \neq 0$), $\frac{\ell}{0} = \infty$ ($\ell \neq 0$), $\frac{\ell}{\infty} = 0$
- ▶ Indeterminate forms: $+\infty - \infty$, $0 \cdot \infty$, $\frac{0}{0}$, $\frac{\infty}{\infty}$

Powers $a_n^{b_n}$:

- ▶ We take the logarithm: $a_n^{b_n} = e^{b_n \log a_n}$
- ▶ Indeterminate forms 1^∞ , 0^0 , ∞^0 (reducible to $0 \cdot \infty$)
- ▶ $0^{+\infty} = 0$ and $0^{-\infty} = +\infty$ are not indeterminate forms

Summary

- ▶ The **algebra of limits** allows us to compute limits of sums, products, quotients and powers
- ▶ **Sign permanence** and the **comparison theorem** relate limits and inequalities
- ▶ The **partial arithmetization** of $\pm\infty$ extends the rules to infinite limits
- ▶ The **indeterminate forms** $(+\infty - \infty, 0 \cdot \infty, \frac{0}{0}, \frac{\infty}{\infty}, 1^\infty, 0^0, \infty^0)$ require a case-by-case study
- ▶ **Taking the logarithm** is the tool for limits of powers $a_n^{b_n}$